Summary in Graph

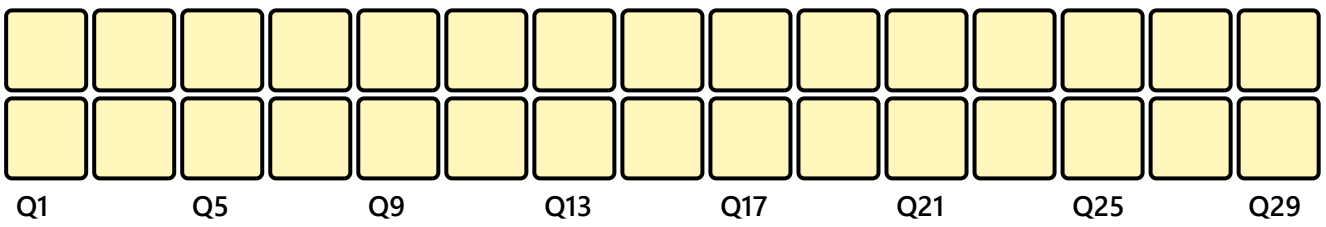
Exam Summary (GO Classes CS Test Series 2026 | Discrete Mathematics | Subject Wise Test 3)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (35) out of 353 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Others

How many paths are there in the plane from $(0,0)$ to $(m,n) \in \mathbb{N} \times \mathbb{N}$, if the possible steps from (i,j) are either $(i+1,j)$ or $(i,j+1)$?

A. $\binom{2m}{n}$
B. $\binom{m}{n}$

C. $\binom{m+n}{n}$
D. m^n

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Let $\mathbb{N} = \{1, 2, 3, \ldots\}$ be the set of natural numbers. For each $n \in \mathbb{N}$, define $A_n = \{(n+1)k,\ k \in \mathbb{N}\}$. Then $A_1 \cap A_2$ equals

A. A_3
B. A_4
C. A_5
D. A_6

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Let A, B and C be three non empty sets. Consider the two relations given below:

$$A - (B - C) = (A - B) \cup C \qquad (1)$$
$$A - (B \cup C) = (A - B) - C \qquad (2)$$

A. Both (1) and (2) are correct
B. (1) is correct but (2) is not
C. (2) is correct but (1) is not
D. Both (1) and (2) are incorrect

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Suppose X and Y are finite sets, each with cardinality n.. The number of bijective functions from X to Y is

A. n^n
B. $n \log_2 n$
C. n^2
D. $n!$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Two sets have m and n elements. The number of subsets of the first set is 96 more than that of the second set. Then the values of m and n are

A. 8 and 6
B. 7 and 6

C. 7 and 5
D. 6 and 5

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Let A be the set of all prime numbers, B be the set of all even prime numbers, and C be the set of all odd prime numbers. Consider the following three statements in this regard:

I. $A = B \cup C$.
II. B is a singleton set
III. $A = C \cup \{2\}$

Then which one of the following holds?

A. None of the above statements is true.
B. Exactly one of the above statements is true.
C. Exactly two of the above statements are true.
D. All the above three statements are true.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Which of the following group contains a self-inverse element other than identity?

A. $(\mathbb{Q}, +)$
B. $(\mathbb{Z}, +)$
C. $(\mathbb{R}, +)$
D. $(\{1, -1, i, -i\}, \times)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

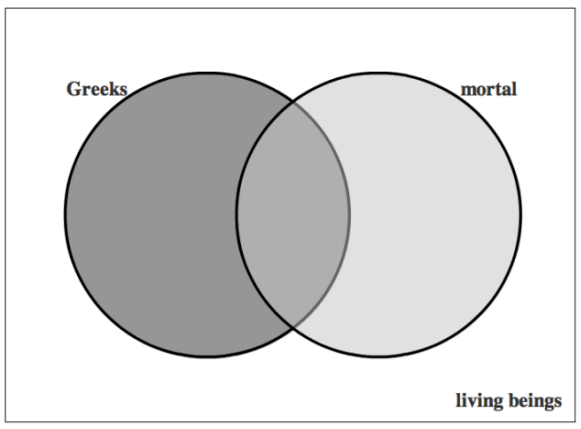
Which formula captures the following statement most accurately?
When the next large bank gets into trouble (t), the financial system collapses (c) unless the Fed buys the bank (b).

A. $(\neg c \rightarrow b) \rightarrow t$
B. $(c \wedge \neg b) \rightarrow \neg t$
C. $(c \wedge \neg b) \rightarrow t$
D. $t \rightarrow (\neg c \rightarrow b)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Consider the following Venn diagram:



Which one of the following categorical propositions can be used in order to express that the dark grey area on the left of the diagram does not contain any elements?

A. Some Greeks are not mortal
B. Some Greeks are mortal
C. All Greeks are mortal
D. No Greeks are mortal

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

What is a bound variable?

A. A variable is bound if and only if it is in the scope of a quantifier that quantifies for some variable.
B. A variable is bound if and only if it is a name.
C. A variable is bound if and only if it is in the scope of a quantifier.
D. A variable is bound if and only if it is in the scope of a quantifier that quantifies for that variable.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Twin primes are pairs of numbers p and $p+2$ such that both are primes—for instance, 5 and 7, 11 and 13, 41 and 43. The Twin Prime Conjecture says that there are infinitely many twin primes.
Let $\text{TwinPrime}(n)$ be a predicate that is true if n and $n+2$ are twin primes. Which of the following formulas, interpreted over positive integers, expresses that there are only finitely many twin primes?

A. $\forall m \cdot \exists n \cdot m \leq n \text{ and not}(\text{TwinPrime}(n))$
B. $\exists m \cdot \forall n \cdot n \leq m \text{ implies TwinPrime}(n)$
C. $\forall m \cdot \exists n \cdot n \leq m \text{ and TwinPrime}(n)$
D. $\exists m \cdot \forall n \cdot \text{TwinPrime}(n) \text{ implies } n \leq m$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Others

Suppose each edge of an undirected graph is coloured using one of three colours — red, blue or green. Consider the following property of such graphs: if any vertex is the endpoint of a red coloured edge, then it is either an endpoint of a blue coloured edge or not an endpoint of any green coloured edge. If a graph G does not satisfy this property, which of the following statements about G are valid?

A. There is a red coloured edge.

B. Any vertex that is the endpoint of a red coloured edge is also the endpoint of a green coloured edge.
C. There is a vertex that is not an endpoint of any blue coloured edge but is an endpoint of a green coloured edge and a red coloured edge.
D. (A) and (C).

Your Answer: | Correct Answer: D;A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

A college prepares its timetable by grouping courses in slots A, B, C, . . . All courses in a slot meet at the same time, and courses in different slots have disjoint timings. Course registration has been completed and the administration now knows which students are registered for each course. If the same student is registered for two courses, the courses must be assigned different slots. The administration is trying to compute the minimum number of slots required to prepare the timetable.
The administration decides to model this as a graph where the nodes are the courses and edges represent pairs of courses with an overlapping audience. In this setting, the graph theoretic question to be answered is:

A. Find a spanning tree with minimum number of edges
B. Find a minimal coloring
C. Find a minimum size vertex cover
D. Find a maximum size independent

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

In a connected undirected graph, the distance between two vertices is the number of edges in the shortest path between them. Suppose we denote by P the following property: *there exists a vertex that is a neighbour of all other vertices*. Consider the following statements:

1. If P is false, then there is a pair of vertices such that the distance between them is at least 4.
2. If P is true, then the distance between any pair of vertices is at most 2.

What can you say about these statements?

1. Only i is true
2. Only ii is true
3. Both i and ii are true
4. Neither i nor ii is true

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

A dodecahedron is a regular solid with 12 faces, each face being a regular pentagon. How many edges are there? And how many vertices?

A. 60 edges and 20 vertices
B. 30 edges and 20 vertices
C. 60 edges and 50 vertices
D. 30 edges and 50 vertices

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Multiple Choice Type Award: 2 Penalty: 0.67 Others

ScamTel has won a state government contract to connect 17 cities by high-speed fibre optic links. Each link will connect a pair of cities so that the entire network is connected-there is a path from each city to every other city. The contract requires the network to remain connected if any single link fails. What is the minimum number of links that ScamTel needs to set up?

A. 34
B. 32
C. 17
D. 16

Your Answer: Correct Answer: C Not Attempted Time taken: 00min 00sec Discuss

Q #17 Multiple Choice Type Award: 2 Penalty: 0.67 Others

Let $G = (V, E)$ be an undirected simple graph, and s be a designated vertex in G. For each $v \in V$, let $d(v)$ be the length of a shortest path between s and v. For an edge (u, v) in G, what can not be the value of $d(u) - d(v)$?

A. 2
B. -1
C. 0
D. 1

Your Answer: Correct Answer: A Not Attempted Time taken: 00min 00sec Discuss

Q #18 Multiple Choice Type Award: 2 Penalty: 0.67 Others

There are 7 switches on a switchboard, some of which are on and some of which are off. In one move, you pick any 2 switches and toggle each of them-if the switch you pick is currently off, you turn it on, if it is on, you turn it off. Your aim is to execute a sequence of moves and turn all 7 switches on. For which of the following initial configurations is this not possible? Each configuration lists the initial positions of the 7 switches in sequence, from switch 1 to switch 7.

A. (off,on,off,on,off,off,on)
B. (off,on,on,on,on,on,off)
C. (on,off,on,on,on,on,on)
D. (off,off,off,off,off,on,off)

Your Answer: Correct Answer: C Not Attempted Time taken: 00min 00sec Discuss

Q #19 Multiple Choice Type Award: 2 Penalty: 0.67 Others

There are five buckets, each of which can be open or closed. An arrangement is a specification of which buckets are open and which bucket are closed. Every person likes some of the arrangements and dislikes the rest. You host a party, and amazingly, no two people on the guest list have the same likes and dislikes. What is the maximum number of guests possible?

A. 5^2
B. $\binom{5}{2}$
C. 2^5
D. 2^{2^5}

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

There are n members of Chennai Mathematical Institute. Most of them are very studious and like to own lots of books. Now the following facts have been learnt.

- No two members own exactly the same number of books.
- Each member owns strictly less than n books.
- No member has exactly 200 books.

Given the above information, which of the following is not a possible value of n?

A. 100

B. 199

C. 200

D. 201

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

How many elements are in the following set?

$$\{(A, B) \mid A \subseteq B \subseteq \{1, 2, 3, \ldots, n\}\}$$

A. 2^{n-1}
B. 3^n
C. 2^{n+1}
D. 2^{2n}

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 01sec | Discuss

Q #22 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Consider all possible words obtained by arranging all the letters of the word $\mathbf{AGAIN}$. These words are now arranged in the alphabetical order, as in a dictionary. The fiftieth word in this arrangement is

A. IAANG
B. NAAGI
C. NAAIG
D. IAAGN

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

If the letters of the word $\mathbf{COMPUTER}$ be arranged in random order, the number of arrangements in which the three vowels O, U and E occur together is

A. $8!$
B. $6!$
C. $3!6!$
D. None of these

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Let X be the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Define the set $\mathcal{R}$ by $\mathcal{R} = \{(x, y) \in X \times X : x$ and y have the same remainder when divided by $3\}$.

Then the number of elements in $\mathcal{R}$ is

A. 40
B. 36
C. 34
D. 33

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Let G be a group with identity element e. If x and y are elements in G satisfying $x^5y^3 = x^8y^5 = e$, then which of the following conditions is true?

A. $x = e,\ y = e$
B. $x \neq e,\ y = e$
C. $x = e,\ y \neq e$
D. $x \neq e,\ y \neq e$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Let G be a finite group of even order. Then which of the following statements is correct?

A. The number of elements of order 2 in G is even
B. The number of elements of order 2 in G is odd
C. G has no subgroup of order 2
D. None of the above.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Numerical Type | Award: 2 | Penalty: 0 | Others

Let $B = \{1, 2, 3, 4\}$. A set $S \subseteq B \times B$ called a symmetric set of B if for all $x, y \in B$,

$$(x, y) \in S \Rightarrow (y, x) \in S.$$

Find the number of symmetric sets of B.

Your Answer: | Correct Answer: 1024 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

If b and c are elements in a group G and if $b^5 = c^3 = e$, where e is the identity of G, then the inverse of $b^2cb^4c^2$ must be:

A. $cb^2c^2b^4$
B. $c^2b^4cb^2$
C. cbc^2b^3
D. $b^4c^2b^2c$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Let $\mathbb{Z}$ be the group of integers under the operation of addition. Which of the following subsets of $\mathbb{Z}$ is not a subgroup of $\mathbb{Z}$?

A. $\mathbb{Z}$
B. $\{n \in \mathbb{Z} : n \geq 0\}$
C. $\{n \in \mathbb{Z} : \mathrm{n} \text{ is even } \}$
D. $\{n \in \mathbb{Z} : 6 \mid n \text{ and } 9 \mid n\}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

A group G in which $(ab)^2 = a^2b^2$ for all a, b in G, is necessarily

A. Abelian
B. Finite
C. Cyclic
D. of order 2

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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